

**First Mid Semester Examination,
Cloud Computing-II CS603(B)
Branch: CSE-Semester: VI**

Time 1:30 Hrs**Max. Marks 20****Q.1**

- a) Define *Continuous Integration (CI)* and *Infrastructure as Code (IaC)* in the context of DevOps.
b) Explain how a *faulty script deployment* during CI can lead to unexpected service downtime.
(CO1 – 5 marks)

OR

- a) Suppose an enterprise is facing delays in deployment due to manual release cycles. As a DevOps engineer, how would you implement *Continuous Delivery (CD)* to address the issue?
b) Compare and contrast *DevOps lifecycle* with *traditional software operations*. Highlight key differences.

Q.2

Evaluate the effectiveness of the *DevOps Maturity Lifecycle* in large-scale software organizations. What improvements would you suggest to accelerate DevOps adoption? (CO1 – 5 marks)

OR

Design a *DevOps onboarding module* for junior developers that highlights key areas such as source control, configuration management, testing, and automation practices.

Q.3

- a) List five essential tools or practices involved in a *DevOps process workflow*.
b) Describe the role of *Artifacts Repository Management* in maintaining versioned and secure software builds. (CO2 – 5 marks)

OR

- a) You are part of a DevOps team. What steps would you take to manage a failing *build pipeline* due to misconfigured code repositories?
b) Analyze the difference between *routine automation* and *test automation* in the DevOps framework. How do they contribute to productivity?

Q.4

Assess the impact of *poor source code and configuration management* in a DevOps environment. Are current frameworks like Git and Jenkins sufficient to address these issues? Justify your answer.
(CO2 – 5 marks)

OR

Create a case study where *inadequate DevOps maturity and poor testing* led to a failed release in production. Suggest a mitigation strategy using best DevOps practices.